

Neural Networks

Seminar - Week 1 - Introduction

Communication

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Recommended Courses & Literature

- **Ian Goodfellow**
 - <https://www.deeplearningbook.org/>
- **Andrew Ng**
 - <https://www.coursera.org/specializations/deep-learning>
- **Dive into Deep Learning**
 - <https://d2l.ai/index.html>

Assessment methods and criteria

Min

Semester		60 points	
	Pop-up quiz	1 + 1 + 1 points	
	Seminar Assignments	2 + 2 + 3 points	2 points
	Projects	8 + 12 + 15 points	15 points
	Midterm Test	15 points	5 points
Final Exam		40 points	10 points
Total		100 points	56 points

Week	Seminar
1	Introduction
2	Assignment - Activation Functions, MLP, Forward Pass
3	Assignment - Loss Functions, Derivatives, Back-propagation
4	Assignment - Optimizers
5	Consultation - Project 1
6	Presentation - Project 1
7	Consultation - Project 2
8	Consultation - Project 2
9	Presentation - Project 2
10	Consultation - Project 3
11	Consultation - Project 3
12	Presentation - Project 3

Seminar assignments

- Coding an MLP from Scratch
 - **Individual work**
 - **Goal:** Better understanding of how neural networks work
 - Finishing existing code
-
- Public GitHub - New assignment every week
- https://github.com/vgg-fiit/NSIETE_2026
-
- Submit to AIS **as compressed ZIP archive** (.ipynb + PDF export)
 - Deadline: 2 days after the seminar

Projects

- **MLP, CNN, NLP, Image synthesis, Timeseries, ...**
 - **Teams of 2 Students**
 - Usage of existing frameworks
 - Maximum 2 groups per dataset
 - Information 1 week before project start
-
- Submit to AIS **as compressed ZIP archive** (all code, no need for data)

What You Need Installed

Python

- <https://www.python.org/>
- venv/conda environment

Libraries

- Some deep learning framework
- numpy
- pandas
- matplotlib
- Pillow/cv2
- ...

Python deep learning frameworks

PyTorch

- <https://pytorch.org/>
- <https://www.learnpytorch.io/>
- <https://pytorch.org/tutorials/>

TensorFlow

- <https://www.tensorflow.org/learn>

We really recommend PyTorch

Computing options (GPU)

Own hardware

- with NVIDIA GPU

Google Colab

- <https://colab.research.google.com/>

Kaggle

- <https://www.kaggle.com/>

MS Azure and other

Google Colab

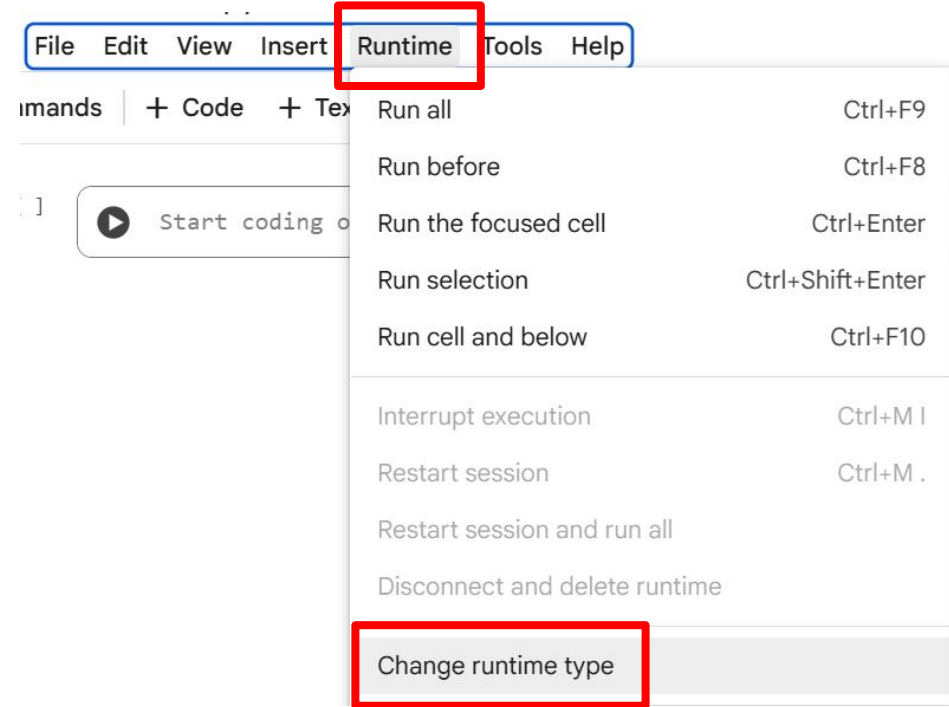
Advantages

- **Free GPU runtime**
- No needed installation
- Pre-installed libraries (most of them)
- Collaboration-friendly

Disadvantages

- **Session time limits**
- Internet dependency
- Environment resets
- Slower with large datasets

Google Colab - Runtime



Change runtime type

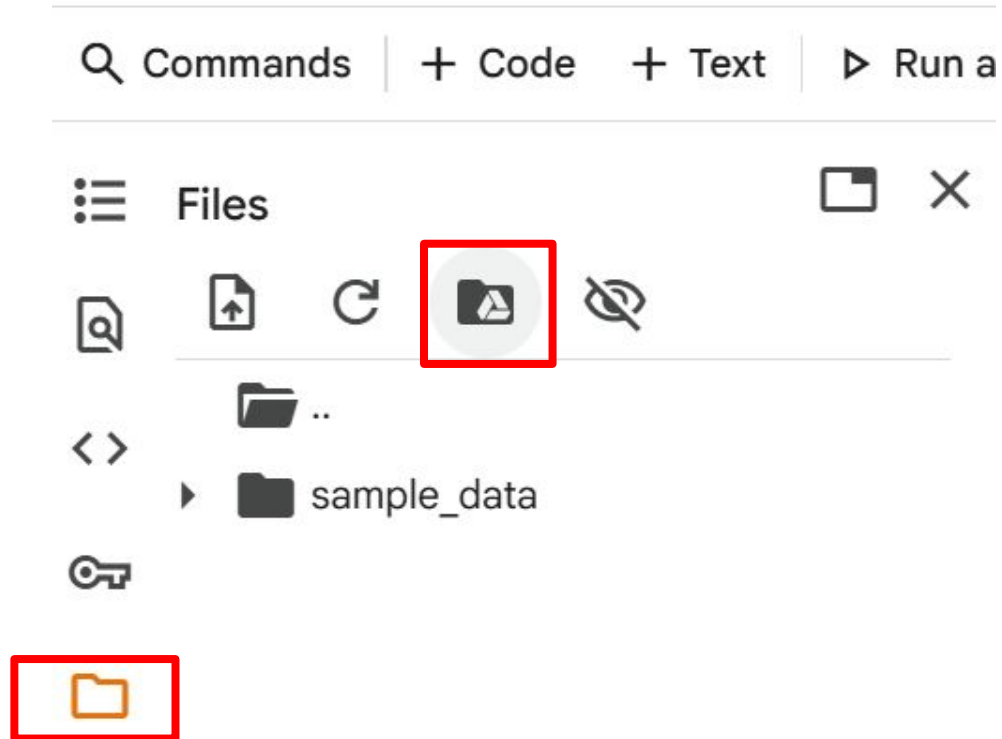
Runtime type

Python 3

Hardware accelerator ?



Google Colab - Mount GDrive



Check if GPU can be used

PyTorch

```
```
```

```
import torch
```

```
Check if CUDA (GPU support) is available
```

```
print("CUDA available:", torch.cuda.is_available())
```

```
Check how many GPUs are accessible
```

```
print("Number of GPUs:", torch.cuda.device_count())
```

```
```
```

Check if GPU can be used

TensorFlow

```
~~~  
  
import tensorflow as tf  
  
# Check if TensorFlow can access GPUs  
print("Num GPUs Available:",  
      len(tf.config.list_physical_devices('GPU')))  
  
# List available GPUs  
print("Available GPUs:", tf.config.list_physical_devices('GPU'))  
~~~
```

Kaggle

Advantages

- **Free GPU runtime (30h per week)**
- No needed installation
- Pre-installed libraries (most of them)

Disadvantages

- **A bit hassle to work with**
- Internet dependency
- You need to “be active” for the session to stay running

Kaggle - Upload dataset and code as dataset (zip)

kaggle

+ Create

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This is the **private view** of your content. To see what others see, visit [Your Profile](#)

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New Model

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New Project Writeup

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Recently Updated

0 selected



GAN_code

Richard Szarka · Updated a year ago
Private · Usability 2.5 · 20 MB



GAN_data

Richard Szarka · Updated a year ago
Private · Usability 1.3 · 112 MB

Kaggle - setup paths for your code and dataset

```
os.chdir('/kaggle/input/gan-code/PG')  
os.getcwd()
```

Change working directory to your project

```
'/kaggle/input/gan-code/PG'
```

```
import torch  
"cuda" if torch.cuda.is_available() else "cpu"
```

Make sure CUDA works properly

```
'cuda'
```

```
!pip install --upgrade -q wandb
```

If using logging tools as wandb, install it

```
from kaggle_secrets import UserSecretsClient  
user_secrets = UserSecretsClient()  
wandb_api = user_secrets.get_secret("wandb_api")
```

You can configure your API key in kaggle secrets

```
import wandb  
wandb.login(key=wandb_api)
```

Login to Wandb

```
wandb: WARNING Path /kaggle/input/gan-code/PG/wandb/ wasn't writable, using system temp directory.  
wandb: WARNING Path /kaggle/input/gan-code/PG/wandb/ wasn't writable, using system temp directory  
wandb: WARNING Path /kaggle/input/gan-code/PG/wandb/ wasn't writable, using system temp directory  
wandb: W&B API key is configured. Use `wandb login --relogin` to force relogin  
wandb: WARNING If you're specifying your api key in code, ensure this code is not shared publicly.  
wandb: WARNING Consider setting the WANDB_API_KEY environment variable, or running `wandb login` from the command line.  
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc  
True
```

```
!python projected_gan.py
```

Run your script

Input

+ Add Input

Upload

DATASETS

- gan-adenoma-data
- gan-code
 - PG
 - check_generator.py
 - config.py
 - dataset.py
 - differentiable_augmentation.py
 - diffusion.py
 - efficient_net.py
 - efficientnet_lite1.pth
 - generator.py
 - projected_gan.py
 - utils.py
- gan-data
 - Coast
 - Desert
 - Forest
 - Glacier
 - Mountain

You can import your uploaded datasets here

Output

- /kaggle/working

Table of contents

Kaggle - customize session

```
[4]: os.chdir('/kaggle/input/gan-code/PG')
     os.getcwd()

[4]: '/kaggle/input/gan-code/PG'

[5]: import torch
     "cuda" if torch.cuda.is_available() else "cpu"

[5]: 'cuda'

[6]: !pip install --upgrade -q wandb

[7]: from kaggle_secrets import UserSecretsClient
     user_secrets = UserSecretsClient()
     wandb_api = user_secrets.get_secret("wandb_api")

[8]: import wandb
     wandb.login(key=wandb_api)

wandb: WARNING Path /kaggle/input/gan-code/PG/wandb/ wasn't writable, using system temp directory.
wandb: WARNING Path /kaggle/input/gan-code/PG/wandb/ wasn't writable, using system temp directory
wandb: WARNING Path /kaggle/input/gan-code/PG/wandb/ wasn't writable, using system temp directory
wandb: WARNING W&B API key is configured. Use "wandb login --relogin" to force relogin
wandb: WARNING If you're specifying your api key in code, ensure this code is not shared publicly.
wandb: WARNING Consider setting the WANDB_API_KEY environment variable, or running "wandb login" from the command line.
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc

[8]: True
```

No sections detected
Add markdown headers to add a section

Session options

ACCELERATOR
GPU T4 ×2

Quota: 00:00 / 30 hrs - [Link to Colab Pro for more Quota](#)

LANGUAGE
Python

PERSISTENCE
No persistence

ENVIRONMENT
Pin to original environment (2024-07-11)

You won't get new packages, but your code is less likely to break: [What is a notebook environment?](#)

INTERNET
☒ Internet on

TAGS
Add Tags

Schedule a notebook to run

Code Help

At the bottom
You can customize
the session options

Week 1 - TASK

- **Prepare and test computing environment**
 - Is GPU available?
- **Learn/Refresh knowledge of Python programming**
- **Introduction to Numpy**

https://github.com/vgg-fiit/NSIETE_2026